

John Pumayalli

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EDUCATION

Stanford University Expected June 2027

BS Symbolic Systems — AI & Robotics Concentration · GPA: 3.61 Stanford, CA

Coursework: Machine Learning, Algorithms, Mathematical Foundations, Probability, Computer Organization, Mathematical Logic, Minds & Machines, Philosophy of AI, Semantics & Pragmatics

Telluride High School June 2023

GPA: 4.34 · *Magna Cum Laude* · *Ronald Reagan Leadership Medal* Telluride, CO

EXPERIENCE

OrgOS June 2025 – Present

Software Engineer Intern Remote

- Built a production Python pipeline ingesting Slack data from NeonDB and using Gemini 2.5 Flash (OpenRouter) to infer org structure — actors, roles, seniority scores, team groupings — across thousands of messages
- Architected GCP deployment with Secret Manager, CI/CD via GitHub Actions, and Terraform-managed infrastructure; validated all LLM outputs as structured JSON before NeonDB write

Stanford Sierra Camp June – Sep 2025

Camp Counselor South Lake Tahoe, CA

- Designed and led daily programming for 300+ weekly guests over a 13-week season; built relationships across a Stanford alumni network

Flourish June – Sep 2022

Front End Developer Dartmouth, NH

- Built patient-facing interface for a chronic-condition health app using React Native and Node.js; constructed MongoDB backend in coordination with the UI/UX team

Colorado School of Mines June – Sep 2021

Research Assistant · *Dr. Chuan Yue* Golden, CO

- Reverse-engineered Grammarly to identify privacy vulnerabilities and cybersecurity issues in the browser extension

PROJECTS

Beyond Pip Counting: Interpretable ML for Catan Placement Stanford CS229

Python · *XGBoost* · *SHAP* · *scikit-learn* · *k-means*

- Lifted win-prediction AUC from 0.54 (pip-only baseline) to 0.73 using 52 engineered features across resource scarcity, turn order, opponent competition, and strategic engines — trained on ~100K simulated games
- Applied SHAP to attribute XGBoost predictions; distilled model into a shallow decision tree where 8 human-readable rules recovered 62% of its edge over pip counting

LEADERSHIP & ACTIVITIES

Sigma Phi Epsilon — Social Chair & Head of Member Development Sep 2024 – Present

Mestiza Qoyllacha — Member & Performer Sep 2012 – Present

Annual collaborator with an Indigenous dance collective in the Sacred Valley of Peru; developed cross-cultural communication and teamwork.

SKILLS

Languages: Python, C++, JavaScript, Java, Swift, HTML/CSS, SQL · Spanish (native), English (native), Italian (intermediate)

ML / AI: XGBoost, scikit-learn, SHAP, LLM APIs, OpenRouter, Gemini 2.5, logistic regression, k-means

Infrastructure: GCP, GitHub Actions, Terraform, PostgreSQL, NeonDB, MongoDB

Frameworks: React, React Native, Node.js, Figma, CAD